



Pyrolysis of a fraction of waste polypropylene and polyethylene for the recovery of BTX aromatics using a fluidized bed reactor

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ABSTRACT

Fractions of waste polypropylene and polyethylene were pyrolyzed in a pyrolysis plant under different conditions. In this study, the influence of the reaction temperature (650–750 °C), the feed rate, and the kind of fluidizing medium on the product spectrum were investigated. Pyrolysis of the PP fraction produced oils up to 43 wt.% of the product. With respect to the PE fraction, the maximum oil yield was above 60 wt.% of the product. The target compound was BTX aromatics, whose amount in the oils reached 53 wt.% for the PP fraction and 32 wt.% for the PE fraction. It was shown that the PE fraction yielded a higher liquid product compared to the PP fraction, and that the concentration of aromatics in the oil increased at higher reaction temperatures for both the PP and PE fractions. A higher feed rate and the use of a gas product as the fluidizing medium were favored for the production of oils for both the PP and PE fractions. The oils that were obtained in the experiments almost had no metal and chlorine contents. The maximum heating value of the gas obtained in the experiments was about 50 MJ/kg.

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1. Introduction

Waste plastics cause considerable problems in the world due to their large volume and the difficulty of treating them. Incineration and landfill, which are traditional treatment methods for waste plastics, cannot solve the problem regarding the treatment of plastic wastes because suitable depots are expensive and because incineration stimulates the emission of harmful gases [1]. Furthermore, waste plastics can only be partly recycled into new products due to their contamination [2].

A promising treatment process for waste plastics is pyrolysis. The thermal and catalytic pyrolysis of waste plastics into fuel-like hydrocarbons or monomer under different conditions has been well investigated by some researchers [3–10]. The pyrolysis of the polyalkene plastics LDPE, HDPE, and PP yields an oil product consisting mainly of alkenes, alkanes, and alkadienes; a gas product consisting of hydrogen, alkane, and alkene; and negligible char [11,12]. Kaminsky et al. pyrolyzed mixed plastic wastes in a fluidized bed reactor for the recovery of aromatics [13]. In the pyrolysis at the reaction temperature of 730 °C,

benzene, toluene, xylenes, and styrene made up 31.4 wt.% of the organic products. Kim et al. carried out the pyrolysis of mixed plastic wastes in a technical-scale fluidized bed reactor [14]. At a reaction temperature of 690 °C, about 40 wt.% of the product was pyrolysis oils, about half of which consisted of benzene, toluene, and xylenes. Simon et al. investigated the influence of the pyrolysis temperature on the olefin yields from mixed plastic wastes, using steam as a fluidizing agent [15]. High amounts of olefins were obtained at temperatures of around 700 °C, with 20–31 wt.% ethene, 14–18 wt.% propene, 3–6 wt.% butanes, and 19–23 wt.% pyrolysis gasoline as the main products.

Many researches on the pyrolysis of waste plastics have been carried out in a fluidized bed reactor. Pyrolysis in a fluidized bed reactor, which offers high heat and mass transfer and consequently almost a constant temperature, provides a more uniform spectrum of products due to the shorter residence times in the reactor. As a result of the uniformity of the product spectrum, the possibility of applying the process at a relatively small scale gives rise to a wider range of investment alternatives [16].

The annual generation of waste plastics in South Korea currently amounts to 4.6 million ton/year, and is continually increasing [17]. Polypropylene and polyethylene are the main components of such plastic wastes [18]. In this study, two fractions of waste plastics, polypropylene and polyethylene, were pyrolyzed in a pyrolysis plant equipped with a fluidized bed reactor and a char removal system. The aim of this study was to find out the optimal conditions for a high

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Table 1
Main properties of the feed materials.

Proximate analysis	PP fraction (wt.%)	PE fraction (wt.%)	Ultimate analysis	PP fraction (wt.%)	PE fraction (wt.%)
Moisture	0.15	0.1	C	80.3	81.8
Volatile matter	95.08	98.87	H	13.2	13.6
Fixed carbon	1.22	0.04	Cl	0.06	0.01
Ash ^a	3.55	0.99			

^a By difference.

yield of BTX aromatics. For this purpose, the effect of the pyrolysis parameters (reaction temperature, feed rate, and the kind of fluidizing medium) on the spectrum of products was investigated.

2. Experiment

2.1. Feed material

Two pellet-type fractions of waste plastics, polypropylene and polyethylene, were obtained from a recycling center in Korea. They were sieved to obtain materials with a diameter of 2–3.35 mm. Both feed materials had a high content of volatile matter, with a value of about 95 wt.% for the PP fraction and about 99 wt.% for the PE fraction. The elemental analysis showed that both the PP and PE fractions consisted mainly of carbon and hydrogen. The chlorine contents of the PP and PE fractions were 600 and 100 ppm, respectively. The main properties of the feed materials are shown in Table 1.

2.2. Pyrolysis plant

The pyrolysis plant consisted mainly of a feeding system, a fluidized bed reactor, a char removal system, a quench system, and a gas-circulating system. The feeding system was equipped with two screw feeders with variable feed rates of 0–8.6 g/min so they would

input uniformly. To avoid defluidization, the process is usually operated at feed rates below 7.0 g/min. The reactor, which was indirectly heated by electricity, was made of a sus-316 steel tube with a height of 300 mm and an inner diameter of 80 mm. The reaction temperature was determined based on the average value taken from the three thermocouples in the reactor. The gas distributor was made of steel, with 49 bending tubes designed to prevent the fluidizing quartz sand (0.4 mm) from falling through the plate. The char removal system was made up of a cyclone and a hot filter designed to capture particles bigger than 10 and 2 μm , respectively. The quench system, which was cooled to -30°C using ethanol, consisted of four glass condensers for the precipitation of oil and of an electrostatic precipitator by which the aerosols were captured. The non-condensable gas was recycled using a compressor, and was used as the fluidizing gas. During the pyrolysis, the non-condensable gas was sampled, using teflon gas bags (2 L) at 10 min intervals, to analyze their composition. The excess gas was burned in a flare stack to regulate the pressure. A schematic diagram of the process is shown in Fig. 1.

2.3. TGA and DTG experiments

Prior to the pyrolysis experiments in the plant, TGA experiments were carried out with the PP and PE fractions under a nitrogen atmosphere, using a thermogravimetric analyzer (TGA 2050, TA Instruments). All the experiments were conducted within the 30–800 $^\circ\text{C}$ temperature range, at the heating rates of 5, 10, and 20 $^\circ\text{C}/\text{min}$. The sample amount in each experiment was 10–20 mg, and the flow rate of nitrogen gas was 100 ml/min. The TGA and DTG curves for the PP and PE fractions are shown in Figs. 2 and 3, respectively. α on the y axis in Fig. 3 can be expressed using initial (m_0), instantaneous (m), and final mass (m_∞) of the sample by: $\alpha = (m_0 - m)/(m_0 - m_\infty)$. The TGA curves of the PP fraction show that a small weight loss took place after 500 $^\circ\text{C}$. This may have resulted from the degradation of other waste plastics. The TGA curves above 600 $^\circ\text{C}$ indicate that the residue

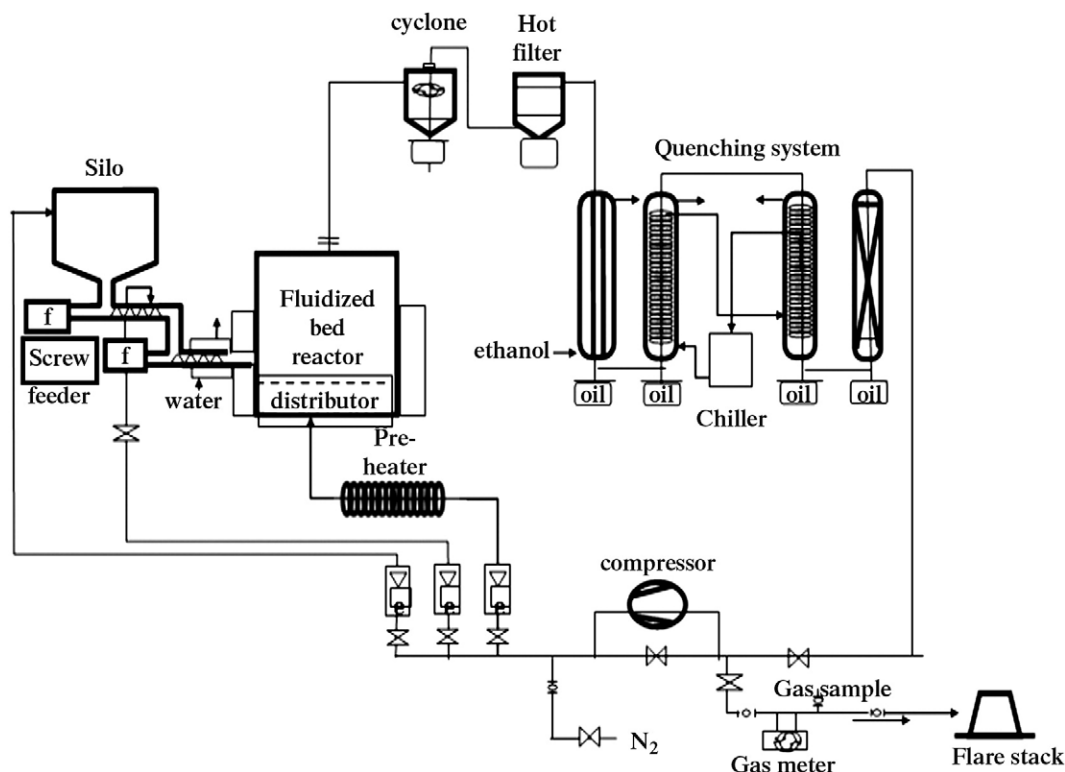


Fig. 1. Schematic diagram of simplified pyrolysis plant (this figure shows the pyrolysis plant applied in the experiments).

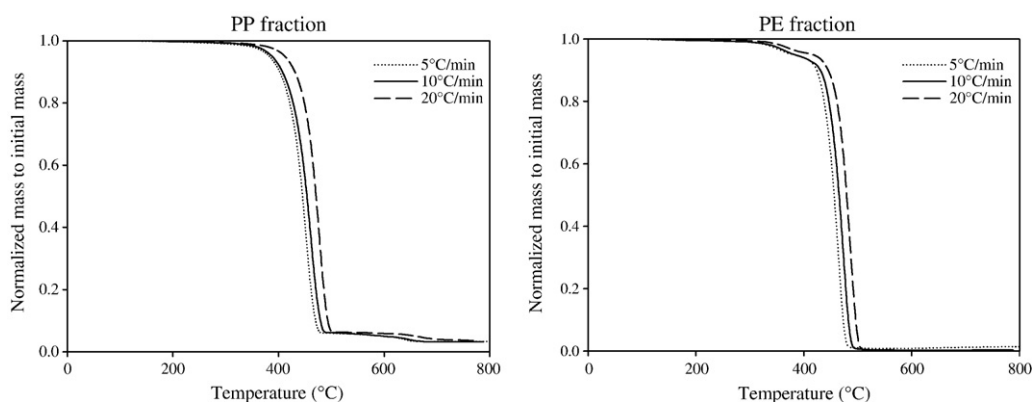


Fig. 2. TGA curves of the PP and PE fraction (the left side is the TGA curve of the PP fraction, and the right one is that of the PE fraction).

weight of the PP fraction was higher than that of the PE fraction. The DTG curves show that the main decomposition occurred within the range of 400–500 °C for both materials and that the weight loss for the PP fraction is more active at a lower temperature (below 400 °C) compared to the PE fraction. The active decomposition of PP is mainly due to the fact that half of the carbons in a PP chain are tertiary carbons and, as such, as a consequence of the ease of formation of tertiary carbocation during the degradation, PP was thermally degraded at a rate faster than that at which PE was thermally degraded [19].

2.4. Reaction conditions

Pyrolysis experiments were conducted to investigate the effects of the main reaction conditions (reaction temperature, feed rate, and the kind of fluidizing medium). In each experiment, the input of the feed material was 0.5 kg, and 1.5 kg quartz sand was used as the fluidized bed material. Except for Run6, where nitrogen gas was applied, a gas product was used as the fluidizing medium. The experiments from Run1 to Run4 for both the PP and PE fractions were performed to determine the influence of the reaction temperature on the aromatics yield. To achieve a high yield of BTX aromatics, reaction temperatures ranging from 660 °C to 750 °C were selected. In each case, Run5 was performed to find out the influence of the feed rate, and Run6 to find out the effect of the kind of fluidizing medium used. The reaction conditions are shown in Table 2. Some of the experiments were performed several times to obtain better results.

2.5. Analysis of the products

The pyrolysis of the two feed fractions produced oil, gas, and solid residues. The three fractions were analyzed using various methods. A

fraction of each oil obtained from each run was distilled in a laboratory scale apparatus under reduced pressure (210 °C, 13.3 kPa) to obtain a light fraction. This distillation condition corresponded to the boiling point of fluorene (bp = 295 °C at atmospheric pressure). After distillation, the oil components, whose boiling points are lower than that of fluorene, were separated from the viscous liquid residue (distillation residue). The light distilled oil was analyzed quantitatively and qualitatively using a GC-MS (HP 5973 inert) and a GC-FID (ACME 6000, Young Lin Instrument Co., Ltd.). The applied capillary column was a Petrocol (100 m × 0.25 mm × 0.5 m), and helium was used as the carrier gas. The metal and chlorine contents of the oils were measured with an ICP-AES (138 Ultrac, Jobin Yvon) and a trace chlorine analyzer (TCL-100, Mitsubishi), respectively. The gas components were analyzed with gas chromatographs (ACME 6000, Young Lin Instrument Co., Ltd.), using a thermal conductivity detector (TCD) and a flame ionization detector (FID), with argon as the carrier gas. The applied columns were Carboxen-1000 (TCD) and HP-plot Al₂O₃/KCl (FID). The quantification of oil and gas components with GCs was carried out with response factor of each component. Elemental analysis (Flash EA 1112 series, CE Instruments) was performed for the liquid and solid products that were obtained in the experiments. To establish organic mass balance in each experiment, the char and distillation residue that were obtained were ignited in a muffle furnace (J-FM2, JISICO) at 850 °C.

3. Results and discussion

3.1. Organic mass balance

In establishing organic mass balance, the product oil and char yields were determined by weighing after the pyrolysis experiments.

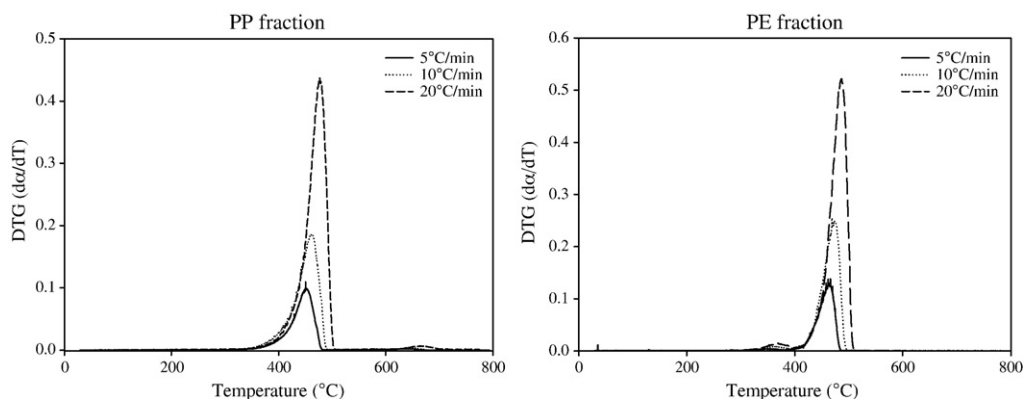


Fig. 3. DTG curves of the PP and PE fraction (the left side is the DTG curve of the PP fraction, and the right one is that of the PE fraction).

Table 2
Reaction conditions.

Parameters	PP fraction						PE fraction					
	Run1	Run2	Run3	Run4	Run5	Run6	Run1	Run2	Run3	Run4	Run5	Run6
Reaction temperature (°C)	668	703	727	746	704	706	660	680	704	728	676	706
Fluidizing medium ^a	P	P	P	P	P	N	P	P	P	P	P	N
Feed rate (g/min)	4.5	4.5	4.5	4.5	2.7	4.5	5.2	5.2	5.2	5.2	2.9	5.2
Flow rate (N/min)	29	28	27	26	28	28	29	28	28	27	28	28

^a P: product gas, N: nitrogen gas.

The amount of gas production was calculated, along with the gas density and the measured gas volume. Thereafter, distillation with the product oil was conducted to separate the light oil from the viscose residue. The light distilled oil and the viscose fraction were designated as oil and distillation residues, respectively. Finally, the char and distillation residues were burnt in a furnace to find out the amount of organic mass in them. The organic mass balance from Run1 to Run4 is shown in Table 3. In the organic mass balance, the gas is composed of up to C₄ hydrocarbons. The higher hydrocarbons than C₄ analyzed in the gas phases was assigned to the oil component.

For the PP fraction, the oil and gas yields were about 30–43 wt.% and 54–66 wt.%, respectively, while for the PE fraction, they were about 38–61 wt.% and about 37–59 wt.%, respectively. The main compounds of the gas product were methane, ethene, ethane, propene, butadiene, and butane. The oil consisted mainly of aliphatics and mono- and polyaromatics. The maximum yield of BTX aromatics

for the PP fraction was obtained at 746 °C, with a value of 15.5 wt.%; for the PE fraction, it was obtained at 728 °C, with a value of 12.4 wt.%. The distillation residue in each of the experiments was very small, with a value of below 2 wt.% for both cases. Compared to the PE fraction, the char production for the PP fraction was higher because of the high ash content in the PP feed material.

3.2. Influence of the reaction parameters on the product distribution

The distribution of all the product fractions with respect to the reaction temperature is given in Fig. 4. The experiments with the waste PP fraction were conducted at 668, 703, 727 and 746 °C, and those with the waste PE fraction at 660, 680, 704 and 728 °C. The exact comparison of both materials at the same temperature could not be accomplished due to the difficulty of temperature control, especially above 720 °C.

Table 3
Organic mass balance of the pyrolysis experiments.

Product	PP fraction (wt.%)				PE fraction (wt.%)			
	Run1 (668 °C)	Run2 (703 °C)	Run3 (727 °C)	Run4 (746 °C)	Run1 (660 °C)	Run2 (680 °C)	Run3 (704 °C)	Run4 (728 °C)
Gases (sum)	54.4	57	61.7	65.9	36.6	42.8	54.6	59.3
Hydrogen	0.6	0.7	0.6	0.01	0.2	0.3	0.4	0.6
Methane	9.8	12.6	16.6	33.9	6.6	8.0	10.1	13.5
Ethene	9.1	11.5	14.7	11.6	11.6	13.7	18.2	21.5
Ethane	5.5	5.8	5.8	7.1	4.1	4.8	5.2	5.2
Propene	15.5	14.5	13.4	5.1	7.8	9.4	11.7	10.5
Propane	1.7	1.3	1	1.3	1	1.1	1.1	0.9
Butane	7.1	5.8	5.1	1.9	0.3	0.4	0.5	0.4
Butene	2.0	1.8	1.7	0.7	2.2	2.5	3.0	2.3
Butadiene	1.4	1.6	1.8	1.1	1.5	1.8	2.6	3
Others	1.7	1.3	1.1	3.1	1.4	0.9	1.7	1.5
Oils (sum)	43.1	35.9	34.7	29.6	61	56.7	43.7	38.2
Aliphatic (sum)	32.8	21.9	13.1	2.9	51.9	42.2	30.1	20.4
2-Methyl-1-Pentene	1	1	0.3	–	–	–	–	–
1-Methylcyclopentene	–	–	1.3	–	0.4	0.3	0.1	0.1
3-Methylcyclopentene	0.7	0.2	0.1	–	0.2	0.1	0.2	0.1
1-Hexene	–	–	–	–	1.6	0.6	0.7	0.5
Cyclohexene	–	0.3	–	–	0.8	0.5	0.6	0.5
1-Heptene	0.3	0.3	0.2	–	1.8	1.2	1.1	0.9
1-Octene	0.4	0.2	0.1	–	1.5	1.3	1	0.8
1-Nonene	–	0.1	–	–	1.7	1.9	1.2	–
1-Decene	0.5	–	–	0.2	2.1	2.2	1.3	1.1
1-Undecene	0.4	–	0.1	–	1.5	1.5	0.9	0.5
1-Tridecene	0.3	–	–	–	–	1	0.6	0.3
Others C4-C13 hydrocarbons	26.7	19.7	11	2.7	27.7	20.6	17.2	13.3
Over C14 hydrocarbons	2.5	0.1	–	–	12.6	11	5.2	2.3
Aromatic (sum)	10.3	14	21.6	26.7	9.1	14.5	13.6	17.8
Benzene	2.6	4.3	7	8.4	2.8	3.9	4.8	6.4
Toluene	3.1	3.8	6.4	5.9	2.4	3.6	3.5	4.4
Xylenes	1.5	1.6	1.9	1.2	0.5	0.8	0.6	1.6
Ethylbenzene	0.4	0.4	0.7	0.8	0.4	0.6	0.5	0.6
Styrene	0.8	0.7	1.4	2.8	0.6	1.1	1	1.5
Indene	0.2	0.3	0.5	1.2	0.4	0.8	0.7	0.8
Biphenyl	–	0.02	0.04	0.2	–	0.03	0.03	0.02
Others	1.7	2.9	3.7	6.2	2	3.7	2.5	2.5
Distillation residue	0.5	0.2	0.5	0.5	1.7	0.2	1.4	0.5
Char	2	6.9	3.1	4	0.7	0.3	0.3	2

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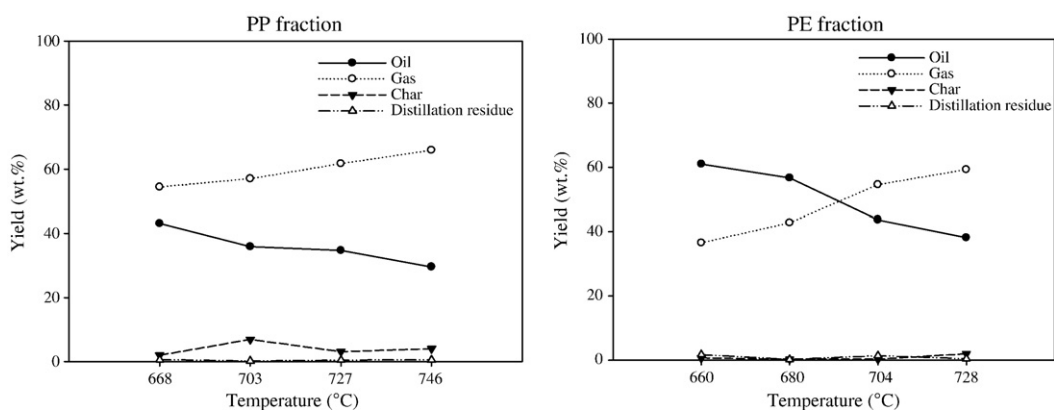


Fig. 4. Product distribution with respect to the reaction temperature (the left side of the figure shows the product distribution according to the reaction temperature with the PP fraction, and the right one is that with the PE fraction).

An increase in the reaction temperature led to a decrease in oil production and an increase in gas production due to the fact that the heavier components were converted into gas components through secondary cracking reactions at elevated temperatures. The amount of gas production summed up to 65.9 wt.% at 746 °C for the PP fraction and to 59.3 wt.% at 728 °C for the PE fraction. The higher amount of gas production for the PP fraction implies the easier degradation of PP compared to PE, due to the intramolecular radical transfer. The amount of oil production decreased from 43.1 to 29.6 wt.% along with the temperature in the case of the PP fraction, and from 61 to 38.2 wt.% in the case of the PE fraction. The result also indicates the difficulty of degrading PE, and the fact that many of the heavier oil components obtained in PE pyrolysis can be converted into light ones, including the gas components, at higher reaction temperatures. The effect of the feed rate on the product spectrum is shown Fig. 5.

As can be seen in Fig. 5, the influence of the feed rate on the oil and gas yields for both the PP and PE fractions (in both cases, the comparison of Run2 and Run5) shows the same trend to give a higher oil yield at a higher feed rate. At a higher feed rate, the yield of oil increased because the higher feed rate caused the higher generation of pyrolysis vapors, which led to the short residence time of the pyrolysis vapors in the reactor and consequently prevented secondary cracking reactions, such as thermal cracking. Fig. 6 shows the influence of the kind of fluidizing medium used on the product distribution.

The left side of the Fig. 6 compares Run2 (product gas) and Run6 (nitrogen) for the PP fraction, and the right one Run3 (product gas) and Run6 (nitrogen) for the PE fraction. In both the cases oil yield increased when the product gas was applied as a fluidizing medium. It seems that nitrogen gas plays a role in diluting the pyrolytic vapors and prevents pyrolytic vapors from reactions leading to the oil

components, such as Diels–Alder reaction that takes place with dienes and alkenes. In contrast to nitrogen, when the product gas was applied, it seems that some of its components reacted with the pyrolysis vapors in the reactor to form oil components.

3.3. Product fraction analysis

Fig. 7 shows the variation of the gas composition with respect to the reaction temperature. For both materials, higher reaction temperatures led to a high methane content that was very stable thermodynamically at those temperatures. In particular, the formation of methane in the PP fraction seemed highly active above 720 °C. Propene was reduced as the reaction temperature rose. In particular, the propene content in the experiments with the PP fraction strongly reduced above 720 °C. It seems that it can be actively used at elevated temperatures in some reactions, such as in the Diels–Alder reaction. The other hydrocarbons content was relatively constant or decreased along with the reaction temperature. The gas product had higher heating values (between 42 and 50 MJ/kg). The gas composition obtained in the experiments was very different from that obtained using steam as a fluidizing medium [15].

Fig. 8 shows the composition of the oils with respect to the reaction temperature. For the PP fraction, the aliphatic concentration in the oil decreased sharply as the reaction temperature went higher. The aliphatic concentration in the oil obtained at 746 °C was only 2.9 wt.%. Meanwhile, for the PE fraction the aliphatic concentration in the oil was 20 wt.% at 728 °C (Table 3). This is another evidence of the difficulty of degrading PE.

The content of BTX aromatics in the oil was 53 wt.% at 746 °C for the PP fraction, and 32 wt.% at 728 °C for the PE fraction. The optimal reaction temperature for the production of BTX aromatics seems to be

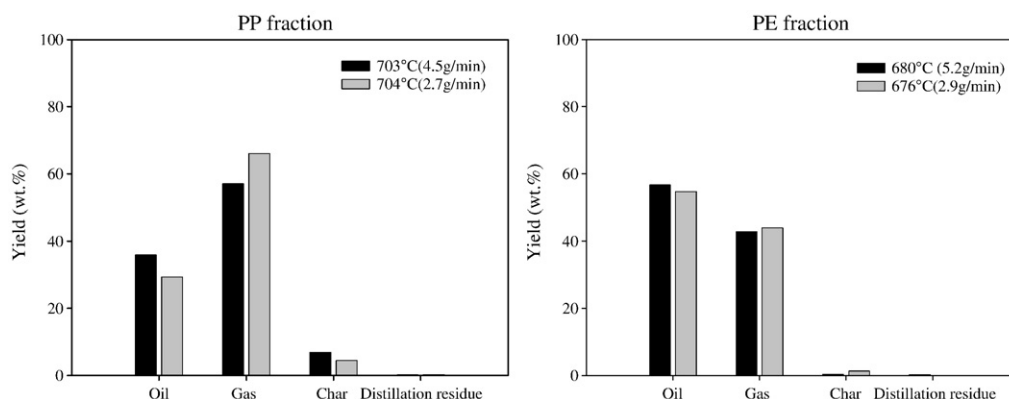


Fig. 5. Effect of the feed rate (the left side of the figure shows the product distribution according to the feed rate with the PP fraction, and the right one is that with the PE fraction).

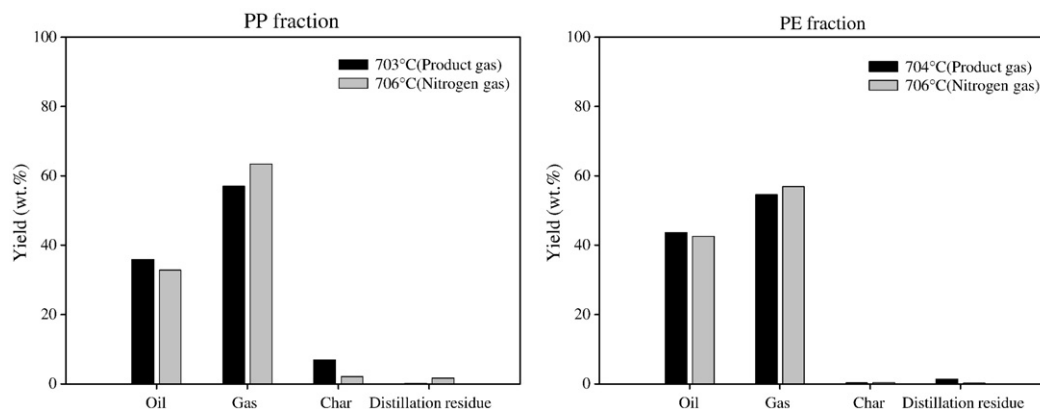


Fig. 6. Effect of the fluidizing medium used (the left side of the figure shows the product distribution according to the kind of fluidizing medium with the PP fraction, and the right one is that with the PE fraction).

at least above 700 °C. Temperatures higher than 700 °C will yield more BTX aromatics. At these temperatures, however, the amount of oil will significantly decrease. The formation of aromatics in the pyrolysis of polyolefin was accomplished using the Diels–Alder reaction, followed by dehydrogenation [20,21]. The detailed mechanism of the formation of BTX aromatics is presented in some papers [22–24]. In general, the aromatic content of the oils for the PP fraction was higher than that for the PE fraction. This may have resulted from the easier formation of propene in the pyrolysis of polypropylene,

which may take part in the Diels–Alder reaction. The easier formation of propene can be explained as follows: the random chain scission of polypropylene produces both primary and secondary radicals, and subsequently, tertiary radicals are formed by intramolecular radical-transfer reactions. The beta-cleavage of the tertiary radicals leads to the easier formation of propene [19]. Fig. 9 shows the distribution of BTX aromatics in the oil with respect to the reaction temperature.

Among the BTX aromatics, in both the PP and PE fractions, benzene was the most abundant compound. The concentration of benzene as

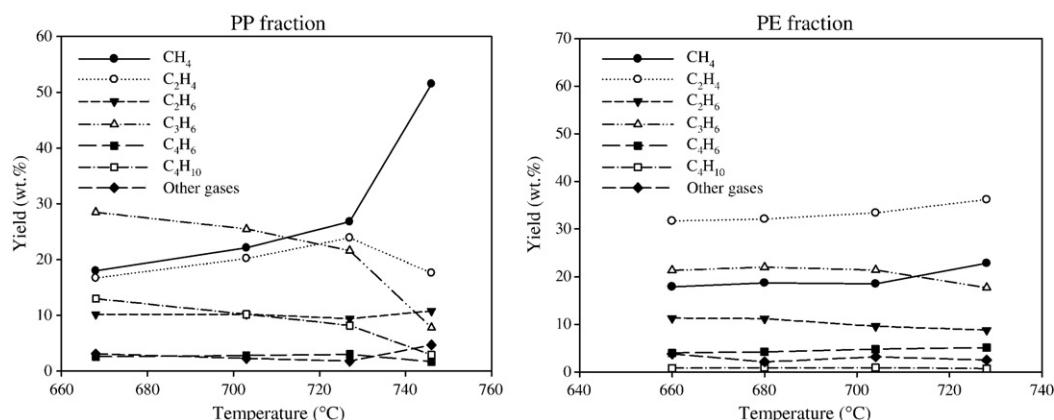


Fig. 7. Development of the gas composition with respect to the reaction temperature (the left side of the figure shows product gas development according to the reaction temperature with the PP fraction, and the right one is that with the PE fraction).

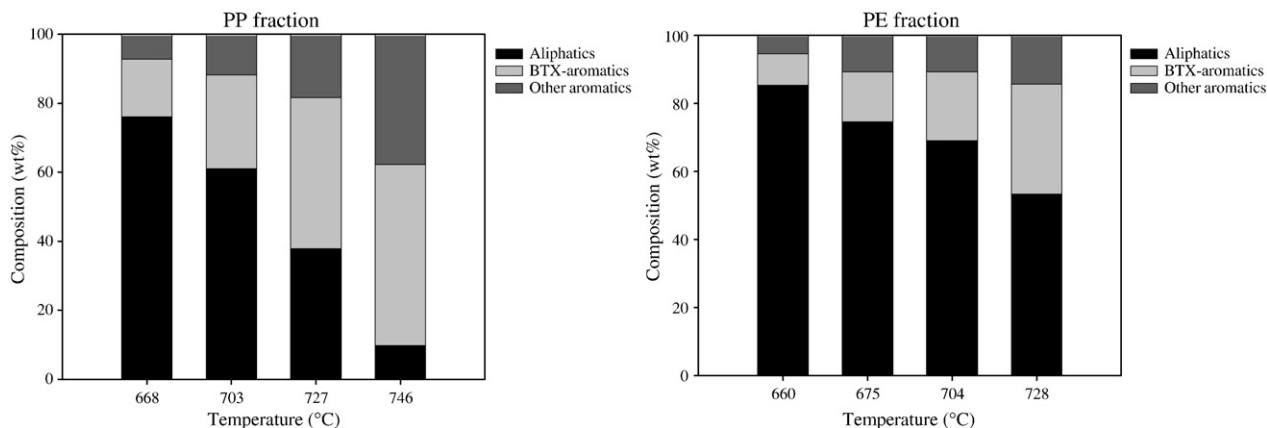


Fig. 8. Development of the oil composition with respect to the reaction temperature (the left side of the figure shows oil composition development according to the reaction temperature with the PP fraction, and the right one is that with the PE fraction).

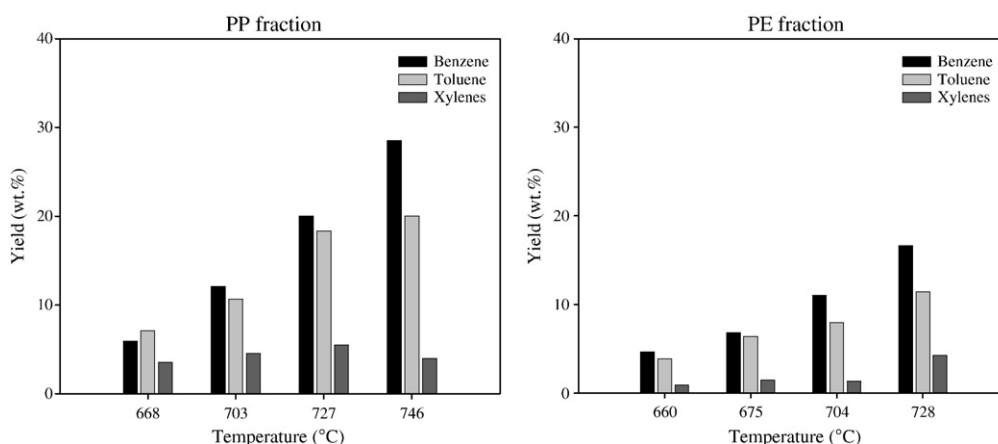


Fig. 9. Distribution of BTX aromatics in the oil with respect to the reaction temperature (the left side of the figure shows the BTX content development according to the reaction temperature with the PP fraction, and the right one is that with the PE fraction).

well as that of toluene increased along with the reaction temperature. The concentration of xylenes did not change significantly with the reaction temperature. Table 4 shows the results of the elemental analysis of the oil.

The maximum C/H ratio of the oil was 0.92 for the PP fraction and 0.66 for the PE fraction, respectively. As can be seen, the C/H ratio of the oil increased as the reaction temperature rose. This indicates that the oil becomes aromatic as the reaction temperature rises.

When metals, such as zinc and lead, are present in the oil, they can react with the oil to form unstable compounds, and they can be liberated into the air when the oil is burnt. Therefore, the oil must contain the least possible metals. During pyrolysis, metals can be carried by the char particles and the pyrolytic vapors. Therefore, char removal before quenching is very crucial to keep the pyrolysis oils clean. To achieve this, the char removal system was applied in the plant. Table 5 represents the results in relation to the metal contents of the pyrolysis products. As can be seen in Table 5, the oil almost has no metal content. The metals are concentrated in the char and in the distillation residue. The metals in the distillation residue may have resulted from the very fine char particles that could not be removed through the char removal system. The efficiency of the char removal system has already been introduced in a paper [25].

In the processing of petrochemicals into valuable chemicals, high chlorine contents are undesirable because of their corrosion potential

[26]. Table 6 shows the chlorine contents of the oils. The amount of chlorine in the oil was below 10 ppm for both the PP and PE fractions. This indicates that the oils obtained in the experiments have good characteristics for utilization as petrochemical feedstock or fuel oil.

3.4. Char and distillation residue analysis

Table 7 shows the elemental analysis of the char and the distillation residue in each experiment. The char from both fractions consisted mainly of inorganic matters that could have originated from the inorganics in the feed fractions and from the fine sand that was used as the fluidized bed material. Due to the high inorganic content of both products, their application as fuels seems difficult. The organic parts of both products show highly aromatized characteristics. In most cases, the C/H ratio of both products is above 1.

3.5. Comparison

The BTX yield in this study can be compared with that of an earlier experiment carried out in Hamburg Process with mixed waste plastics [22]. The mixed waste plastics were mainly composed of polyolefin (about 65%) and polystyrene (about 14%). Because the main component of the mixed waste plastics is polyolefin, the comparison seems to be possible. Fig. 10 shows the results of the comparison

Table 4
Elemental analysis of the oils.

Element	PP fraction (wt.%)						PE fraction (wt.%)					
	Run1	Run2	Run3	Run4	Run5	Run6	Run1	Run2	Run3	Run4	Run5	Run6
C	85.7	86.7	89.1	91.5	88	88.3	85.5	86.2	85.5	87.2	85.9	85.2
H	12.1	10.4	9.6	8.3	10.2	10	13.3	12.5	11.7	11	12.3	13.4
C/H ratio	0.59	0.69	0.78	0.92	0.72	0.74	0.54	0.58	0.61	0.66	0.58	0.53

Table 5
Metal contents of the pyrolysis products.

Metal	PP fraction (ppm)							PE fraction (ppm)						
	Feed	Oil		Char		Distillation residue		Feed	Oil		Char		Distillation residue	
		Run1	Run4	Run1	Run4	Run1	Run4		Run1	Run3	Run1	Run3	Run1	Run3
Al	537.6	9.2	14.6	43,200	40,940	7655	7.4	604.6	30.1	31.1	58,700	64,400	2553	1674
Ba	153.4	1	1	1582	2328	1261	1	6.8	1	1	798.2	981.6	30.3	22.5
Cr	14.4	1	1	811.8	221.5	456.9	1	4.2	1	1	534.5	505.6	23.8	23
Zn	1871.7	0.4	0.04	1863	197	2678	643	65.2	11.8	0.04	1652	993	292.6	125
Pb	35.4	0.02	0.02	2678	265.7	7.6	0.3	18	0.1	0.02	647	236.8	156	70.7

Table 6

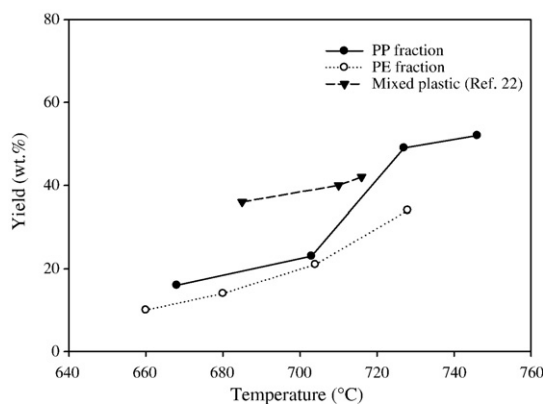
Chlorine contents of the oils.

	PP fraction (ppm)			PE fraction (ppm)		
	Feed	Run1	Run4	Feed	Run1	Run2
Chlorine content	600	4.4	7.6	100	1.3	2

Table 7

Elemental analysis of the char and distillation residue.

Element	PP fraction (wt.%)				PE fraction (wt.%)			
	Run1	Run2	Run3	Run4	Run1	Run2	Run3	Run4
<i>Char</i>								
C	8.2	6.3	11.1	9.8	8	1	4.2	5.2
H	0.2	0.2	0.2	0.3	0.7	0.1	0.2	0.2
Inorganics ^a	91.6	93.5	88.7	89.9	91.4	98.9	95.6	94.6
C/H ratio	3.42	2.63	4.63	2.72	0.95	0.83	1.75	2.17
<i>Distillation residue</i>								
C	66.5	64.8	72.5	64.6	84.8	69.8	83.8	86.1
H	1.7	2.1	3	2.8	10.2	1.9	10	2.9
Inorganics ^a	31.8	33.1	24.5	32.6	5	28.3	6.2	11
C/H ratio	3.26	2.57	2.01	1.92	0.69	3.06	0.7	2.47

^a By difference.**Fig. 10.** Comparison of BTX yields in oil (the figure compares the experimental results with an earlier experiment).

regarding BTX content in oils according to reaction temperature. Although there was a little difference in the feed material composition in both experiments, the content of BTX aromatics increased with reaction temperature. Among the three feed materials, the PP fraction produced BTX aromatics the best above 720 °C. The PE fraction yielded the lowest amount of BTX aromatics among them. In addition, Berruoco et al. studied the influence of temperature on the distribution of liquid products obtained in the pyrolysis of HDPE in a bubbling fluidized bed. In the work, BTEX aromatics made up about 7 wt.% of the products [27].

4. Conclusions

Pyrolysis of polyolefin wastes (PP and PE fractions) was carried out in a fluidized bed reactor equipped with a char removal system, under various reaction conditions (temperature, feed rate, and fluidizing medium). The oil that was obtained in the experiments consisted of aliphatics and mono- and polyaromatic compounds. The content of BTX aromatics in the oil was 53 wt.% at 746 °C for the PP fraction, and

32 wt.% at 728 °C for the PE fraction. For both feed materials, it was found that the concentration of BTX aromatics in the oil increased along with the reaction temperature. In a study that was conducted on the influence of the feed rate and the kind of fluidizing medium used on the oil production, it was found that a higher feed rate and the use of a product gas as the fluidizing medium are more effective for the production of oil. The oils that were obtained in the experiments almost had no metal content. The metals were concentrated in the char and in the distillation residue. In addition, the amount of chlorine in the oil was below 10 ppm for both the PP and PE fractions. Due to the lower metal and chlorine contents of the oil, it seems that the environmental problems caused by the burning of the oil would not be serious. The main compounds of the pyrolysis gas were methane, ethene, ethane, propene, butadiene, and butane, and the maximum heating value of the gas obtained in both cases was about 50 MJ/kg. Therefore, the pyrolysis gas seems to be appropriate for use as a heating source in a pyrolysis plant. Ethene and propene, if separated from the other gas components, can be utilized as chemical feedstock for the production of polyolefin. The char and distillation residue produced in the experiments consisted mainly of inorganics, whose utilization remains a challenge. Pyrolysis char, however, has potential for use in road surfacing and as a building material.

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